

ZEWEI REN

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Summary: Statistics Ph.D. candidate at the University of Missouri expecting to graduate in Dec. 2026, with research focusing on survival analysis and longitudinal data analysis. Completed a summer biostatistics internship at Novartis, contributing to statistical methodology to inform clinical trial design, and currently works as a research assistant providing statistical consulting support to faculty and graduate students through the Center for Applied Statistics and Data Analysis (CASDA). Proficient in R, SAS, and Python, with a strong interest in solving complex real-world problems through appropriate statistical modeling and interdisciplinary collaboration.

EDUCATION

University of Missouri, Columbia, MO
Ph.D. in Statistics

Aug. 2021 – Expected Dec. 2026

University of Connecticut, Storrs, CT
M.S. in Statistics

Aug. 2019 – Jun. 2021

Central South University, Changsha, China
B.S. in Statistics

Sep. 2014 – Jun. 2018

TECHNICAL SKILLS

Programming: R (Markdown/Shiny), SAS, Python, JAGS, SQL (MySQL)

Statistical Methods: Survival Analysis, Longitudinal Data Analysis, Multi-State Models, Frailty Models, Mixture Models, High-Dimensional Data Analysis, Transfer Learning, Bayesian Methods, Causal Inference

Tools: L^AT_EX, Microsoft Office (Excel, Word, PowerPoint), SLURM/HPC Parallel Computing, Nextflow, Codex CLI/IDE(VS Code/Cursor), Git/GitHub

WORK EXPERIENCE

Research Assistant

Aug. 2022 – Present

Center for Applied Statistics and Data Analysis (CASDA), University of Missouri

Columbia, MO

- Consulted on over 40 interdisciplinary research projects with graduate students and faculty, translating research questions into statistical analysis plans covering study design, model selection, and results interpretation.
- Developed and implemented statistical analyses in R, Python, and SAS using a broad range of methods, including experimental design, hypothesis testing, ANOVA, generalized linear models, mixed-effects models, and more advanced statistical models, supporting dissertations, manuscripts, and grant proposals across multiple disciplines.
- Provided R programming support through individual consultations and office hours and delivered statistical computing workshops, including the R workshop for ASA DataFest at Mizzou.

Biostatistician – Summer Intern

Jun. 2025 – Aug. 2025

Novartis

East Hanover, NJ

- Analyzed pooled Inclisiran clinical trial data to evaluate treatment effects on patients' transitions across LDL-C thresholds by guidelines, accounting for interval-censored transition times arising from periodic study visits.
- Implemented time-inhomogeneous interval-censored multi-state models with transition-specific proportional hazards and piecewise-constant baseline intensity approximation to capture repeated transitions above and below LDL-C thresholds, beyond the single-transition framework of conventional Cox models.
- Designed simulation studies comparing the interval-censored multi-state approach with Cox and multi-state models ignoring interval censoring, demonstrating reduced treatment-effect bias as transition rates or observation intervals increased and informing observation-interval selection to balance estimation accuracy with patient visit burden.

Statistician

School of Medicine, University of Missouri

Jan. 2022 – Aug. 2022

Columbia, MO

- Supported over 10 MU Health Care research projects by collaborating with physicians and conducting statistical analysis.
- Improved the automation of R workflows for routine statistical analyses, enabling an end-to-end process from data import and cleaning through statistical analysis to result table generation.
- Reviewed domain-specific literature, primarily in orthopedics, and presented to the biostatistics group to strengthen understanding of the scientific context underlying each analysis.

Data Analyst Intern

Strategical Investment Department, Cameron Pace Group, Beijing Branch

Jan. 2019 – Feb. 2019

Beijing, China

- Collected, visualized, and analyzed data on mergers, acquisitions, and equity investments in the Technology, Media, and Telecommunications(TMT) and glass-free 3D film industries to support investment decision-making.

RESEARCH EXPERIENCE

Subgroup Analysis for Interval-Censored Data with a Cure Fraction by a Mixture Promotion Time Cure Model

Feb. 2024 – Apr. 2026

Working Paper, Ready for Submission; Advisor: Dr. Jianguo Sun

University of Missouri

- Proposed a novel Mixture Promotion Time Cure Model (MPTCM) to identify subgroups with heterogeneous cure rates and survival outcomes.
- Developed a computationally efficient EM algorithm for proposed model, and a variance estimation procedure by profile likelihood with delta method to quantify the model parameters' uncertainty.

Variable Selection and Regression Analysis for Clustered Mixed Panel Count Data with L_0 Seamless Penalty

Jan. 2025 – Present

Working Paper, Complete Draft; Advisor: Dr. Jianguo Sun

University of Missouri

- Developed a unified modeling framework for clustered mixed panel count data that accommodates recurrent event counts and simultaneously panel binary outcomes, while addressing within-cluster correlation. Besides, integrated variable selection and regression modeling using the minimum information criterion (MIC), a seamless L_0 penalty, to identify relevant covariates without the need of parameter tuning.

Instrumental Variable Estimation of Complier Causal Treatment Effect for Interval-Censored Data with a Cure Fraction

Jun. 2026 – Present

Working Paper; Advisor: Dr. Jianguo Sun

University of Missouri

- Proposed a novel causal PTCM to simultaneously estimate causal effect of treatment on survival outcomes and cure probabilities among compliers with interval-censored data.

Application of Multi-state model with Interval Censoring on Longitudinal Data for Optimal Observation Interval Selection in Trial Design Phase

Jun. 2025 – Sep. 2025

Summer intern project; Mentor: Dr. Jahangir Alam

Novartis

- Implemented a time-inhomogeneous interval-censored multi-state model with piecewise-constant baseline hazard approximation, demonstrating through simulation and Longitudinal trial data that it improves estimation accuracy and guides optimal observation interval in clinical trials design.

Bayesian Hierarchical Modeling for Longitudinal PBC Data

Oct. 2022 – Dec. 2022

Course Project, Bayesian Analysis

University of Missouri

- Applied Bayesian hierarchical longitudinal modeling using R/JAGS to analyze Mayo Clinic PBC data, comparing fixed-effect, random-intercept, and random-slope models with frequentist alternatives and estimating patient-specific bilirubin trajectories via MCMC.

Experiment Design & Efficacy Analysis (Phase II)

Aug. 2020

Mentor: Dr. Shoubhik Mondal

UConn Biopharmaceutical Summer Academy with Boehringer Ingelheim

- Calculated necessary sample size using Hypothesis Testing and MCP-Mod method given assumed effect size.
- Used fisher exact test to analyze significant differences of dose effect, and applied ANCOVA model and Logistic Model in detecting significant effects of baseline characteristics on primary and secondary endpoints respectively.

PUBLICATIONS

Sun, Jianguo, Yuxiang Wu, Mingyue Du, and **Zewei Ren**. "A Mixture Model Approach for Subgroup Analysis of Interval-Censored Failure Time Data." *Available at SSRN 5244196*.

Pérez-López, Dennis O., Matthew J. Burke, Chady H. Hakim, James A. Teixeira, Jin Han, Yongping Yue, **Zewei Ren** et al. "Circulatory CCL2 distinguishes Duchenne muscular dystrophy dogs." *Disease Models & Mechanisms* 18, no. 3 (2025).

Anbari, Allison B., Knoo Lee, Jennifer M. Hulett, Lada Micheas, **Zewei Ren**, Anthony Belenchia, and Suhwon Lee. "Exploring disparities in post-cancer treatment instructions: an analysis of rural vs. urban breast cancer survivors in Missouri using BRFSS data." *BMC Health Services Research* 24, no. 1 (2024): 1-10.